

Use of System Dynamics for Studying a Restructured West African Power Pool

M. Gebremicael, H. Yuan and K. Tomsovic

Abstract— *Our research group in this project has been developing models to understand the long-term interactions between investment and performance in the electric power system. We discuss some results and simple models in applying these tools to understand the expected pattern of investment in a proposed West African Power Pool (WAPP). Results show that the interconnection between countries has a clear impact on the local system prices and investments in new construction but there will still be large regional variations in prices and new construction. We consider impacts on regional transmission investment.*

Index Terms— *Deregulation, investor behavior, market models, power system planning, system dynamics.*

I. INTRODUCTION

Deregulation of the electricity sector has proven to be frequently disappointing. In California, the electricity markets opened for business in 1998 and within two short years experienced rotating outages and price spikes at multiples of a 100 or more. One key to the stabilization of the market was the rapid increase in new power plants. The markets worked by creating new investment in generation but it came at the cost of an initial crisis and a later glut of supply. Many claim this instability as a fundamental artifact of poorly designed wholesale electricity markets [1] while others believe greater public investment can “break the cycle of boom and bust” [2]. Governments throughout the developing world have also begun experimenting with deregulation of the electricity sector. Given that crises have arisen in countries with sophisticated infrastructure systems, the concern is that deregulation will have even more dire consequence in the developing world systems with chronic under investment.

With support from NSF, we have been investigating developing models to understand the long-term interactions between investment and performance in the electric power system. Specifically, we have developed modeling and

analysis tools for:

- pricing regimes [3], market transparency [4], and bidding activity under transmission constrained systems [5], in order to understand some of the market forces on suppliers,
- investor behavior, in order to understand the sluggish behavior of investors whose construction of new power plants lags behind the growth in demand [6],
- transmission network planning and its impact on investment decisions in different supply options [7].

In this presentation, we overview some results of applying these tools to understand the expected pattern of investment in the proposed West African Power Pool (WAPP). The main result shows that the interconnection between countries has a clear impact on the local system prices and investments in new construction but there will still be large regional variations in prices and new construction. This work was first reported in [8] where more details can be found with the exception of the newly added sections on transmission investment.

II. BACKGROUND

A. System Dynamic Studies for Power System Planning

Planning models in the electric industry have historically been static. The studies tend to study a fixed set of future possible scenarios for some fixed future time (or a few such time points for multi-stage planning). We are developing models based on market conditions in order to represent the planning process under feedback and dynamic conditions. The modeling is based on the field of System Dynamics (SD), a simulation method pioneered by Forrester [9] and popularized in texts by Ford [10] and Sterman [11]. SD can be defined as [12]

branch of control theory which deals with socio-economic systems and that branch of management science which deals with problems of controllability.

SD differs from the typical detailed modeling approaches of engineering planning studies. The emphasis is on information feedback and general system response to events rather than precise predictions of performance. For example, Fig. 1 shows a Stella[®] model of a simple first order system representing peak demand with exponential growth. Fig. 2

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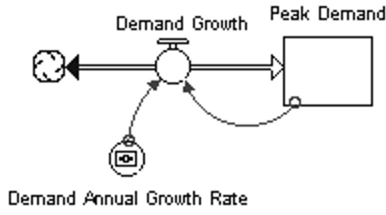


Fig. 1 An example of modeling using exponential growth system dynamics approach

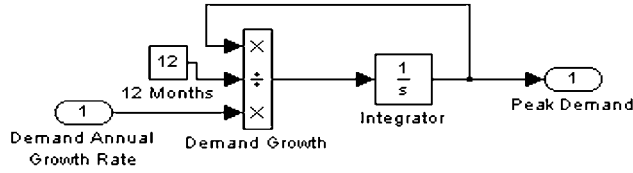


Fig. 2 An example of modeling using an engineering approach (implemented in Simulink)

shows a similar model one might create in an engineering environment, such as Simulink®. The key to practical development of these models is to include all the relevant influences and verifying with historical data.

The traditional engineering approach to modeling arises from an explicit mathematical description of the relations among the system variables. The engineering approach has an advantage over SD with algebraic constraints due to the explicit representation that allows for optimization. In our models, the detailed power flow equations and optimal dispatch needed to determine network flows are modeled by very high level flow models with little explicit network representation. This still allows one to observe the impact of market conditions on planning decisions. For more details see [7, 13-14].

B. Modeling of LMP and Transmission Investment

LMP (Locational Marginal Price) or nodal price is the cost to serve the next MWh of load at a specific location, using the lowest bidding cost of all available generation, while observing all transmission limits. It is a method of determining nodal prices in which market clearing prices are calculated in the competitive wholesale electricity markets. Nodal price theory was first formulated by Schweppe, et al [15], and a general LMP formulation and calculations are given in [16]. LMP can be decomposed into three components [17]. Although the decomposition is a mathematical artifice rather than a physically meaning reality, it still helps us to better understand the LMP and contribute to the market management.

$$LMP = LMP_{energy} + LMP_{congestion} + LMP_{losses} \quad (1)$$

where

LMP_{energy} : Marginal cost of energy

$LMP_{congestion}$: Marginal cost of congestion

LMP_{losses} : Marginal cost of losses

In this research, the system is linearized and transmission losses are ignored. Consider a standard formulation of the DC-OPF

$$\begin{aligned} \min \quad & C^T \underline{P} \\ \text{s.t.} \quad & \sum_{i=1}^m P_i = \sum_{i=1}^m P_{D,i} \quad : \quad \lambda \\ & \underline{P}_L = SF(A\underline{P} - B\underline{P}_D) \leq \underline{P}_L^{\max} \quad : \quad \pi^+ \\ & -\underline{P}_L = -SF(A\underline{P} - B\underline{P}_D) \leq \underline{P}_L^{\max} \quad : \quad \pi^- \\ & \underline{P}^{\min} \leq \underline{P} \leq \underline{P}^{\max} \end{aligned} \quad (2)$$

where,

C : Generator bidding price vector

$\underline{P}$: Generator output vector

P_i : Generator output on bus i

$P_{D,i}$: Load level on bus i

$\underline{P}_L$: Line flow vector

$\underline{P}_L^{\max}$: Line flow limit vector

$\underline{P}^{\min}$: Generator output lower limit vector

$\underline{P}^{\max}$: Generator output upper limit vector

A : Bus - Unit incidence matrix

B : load - incidence matrix

SF : Shift Factor matrix

m : # of buses

λ : dual variable for the power equality constraint

π^+ : dual variable for the line flow constraint in the reference direction

π^- : dual variable for the line flow constraint in the opposite reference direction

For a DC load flow model, losses are ignored so the LMP and its decomposition can be calculated by

$$LMP = LMP_{energy} + LMP_{congestion} = \lambda - SF^T(\pi^+ - \pi^-) \quad (3)$$

Where

$$LMP_{energy} = \lambda \quad (4)$$

$$LMP_{congestion} = -SF^T(\pi^+ - \pi^-) \quad (5)$$

Before the electric industry restructuring, the function of transmission systems is to link generation to load and enhance system reliability so that electric power can be safely, reliably

and economically provided to consumers. With industry restructuring, the role of transmission systems becomes more important. In addition to the historical functions, the functions of transmission systems have been expanded to enhance competition and mitigate market power. With this change, the investment on transmission systems aims not only at maintaining or enhancing system reliability, and linking generation/load to the system, but also at alleviating system congestions, enhancing competition and mitigating market power.

III. WEST AFRICAN POWER POOL

The proposed West Africa Power Pool (WAPP) system is depicted in Fig. 3. We have been adapting our WECC models to the WAPP. Each of the countries in the WAPP suffers from varying degrees of under investment. Not only is there insufficient generation but transmission ties within country and proposed ties between countries tend to be weak. For systems in developing countries (such as in our original work on the WECC), there is far less difficulty in arriving at meaningful parameters for a model. For the WAPP, this problem is not trivial as even the expected demand is difficult to forecast due to underserved load.

Some of the parameters needed for each area include:

- combined cycle (CC) plants total levelized cost,
- construction permit shelf life,
- the goal for permits by developers,
- initial peak annual demand,
- demand annual growth rate,
- investor weight given to CCs in the construction

pipeline,

- generating capacity from all units,
- variable cost of different units,
- natural gas prices,
- transmission network topology and line parameters,
- peak demand for each area over the study period,
- a typical 24 hour demand curve for each month, and
- approximate transmission costs.

A market simulator determines expected prices for typical days during each month of the year and the associated locational marginal prices (LMPs). From these simulated prices, the likely investment in new generation for new CC plants is determined. For new transmission, investment is based on the LMP differences and expected flows. Generally, we assume rational investor behavior.

A. Transmission Planning

In forming competitive electric markets, the originally vertically-integrated and centralized electric industry is restructured to a non-integrated and decentralized industry. Under this change, the traditional transmission planning centralized decision framework is no longer adequate. Before restructuring, the planning process included only the regulators and utilities, which can be fully overseen, but today the planning process includes the regulators, various market players and other interested parties. These inter-relationships cannot be centrally managed. As such, the planning process must move to a more decentralized structure. Moreover, the traditional strict physical models alone do not fully capture this complicated process. The SD approach provides one method to address these complexities.

An SD model can include the simple models of the decisions made by the various market players. Based on

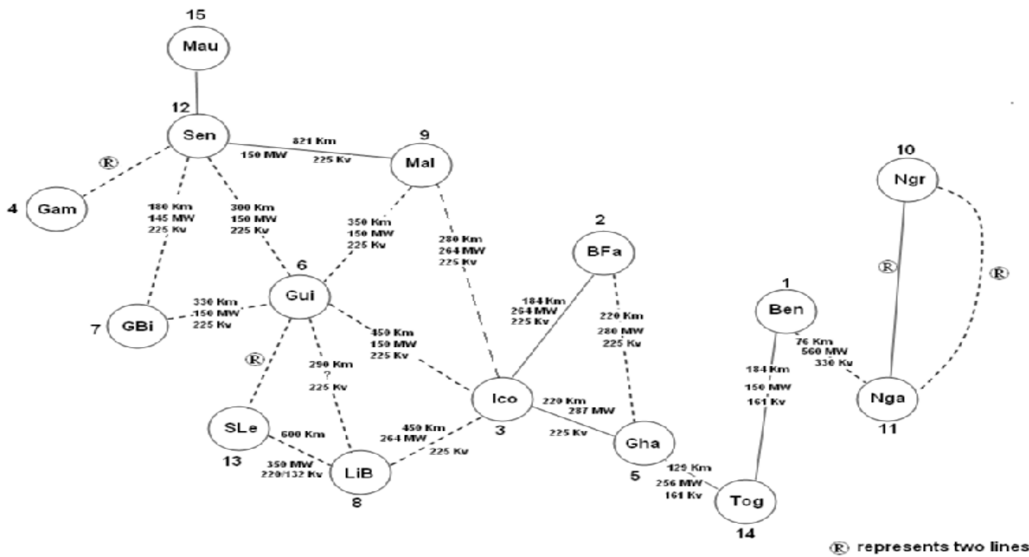


Fig. 3. Interconnections between countries of the WAPP

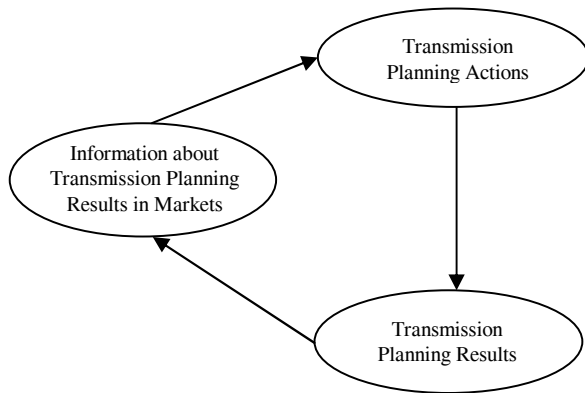


Fig.4 Dynamic transmission planning process

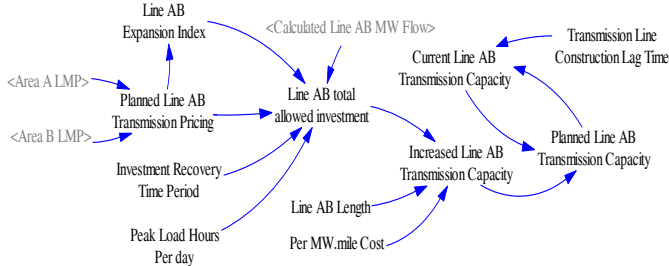


Fig.5 SD model for transmission expansion
(implemented in Vensim)

simulations over the planning horizon, the effects of these uncertainties on transmission expansion can be observed. Moreover the model is closed-loop. Any decision in the previous time and its effects will be considered in the future. Fig. 4 shows this process. We also show a simple SD model in Fig.5 for transmission expansion under incentives from an LMP scheme.

IV. CONCLUSIONS

This presentation describes a simple framework for the study of the construction patterns in the WAPP. The developed model faces several challenges in developing more meaningful results. Namely:

- obtaining more realistic data on costs by country,
- developing a model of investment appropriate for West Africa,
- researching issues associated with under served demand, and
- incorporating other demand side models into the analysis.

V. REFERENCES

- [1] A. Ford, "Boom & Bust in Power Plant Construction: Lessons from the California Electricity Crisis", to appear in a special issue of the *Journal of Industry, Competition and Trade*.
- [2] S. David Freeman, Chairman of the Board of the Power Authority, interviewed in the *Contra Costa Times*, August 13, 2001.
- [3] S. Vucetic, K. Tomsovic and Z. Obradovic, "Discovering Price-Load Relationships in California's Electricity Market," *IEEE Transactions on Power Systems*, Vol. 16, No. 2, May 2001, pp. 280-286.
- [4] L. Xu, K. Tomsovic and A. Bose, "Topology Error Identification using a Two-State DC State Estimator," submitted to *IEEE Transactions on Power Systems*.

- [5] T. Peng and K. Tomsovic, "Congestion Influence on Bidding Strategies in an Electricity Market," *IEEE Transactions on Power Systems*, Vol. 18, No. 3, August 2003, pp. 1054-1061.
- [6] A. Ford, "Simulation Scenarios for the Western Electricity Market: A Discussion Paper for the CEC Workshop on Alternative Market Structures for California," Nov 2001, online at <http://www.wsu.edu/~forda>.
- [7] A. Dimitrovski, K. Tomsovic, and A. Ford, "Comprehensive Long Term Modeling of the Dynamics of Investment and Network Planning in Electric Power Systems," *International Journal of Critical Infrastructures*, Vol. 3, No. 1/2, 2007, pp. 235-264.
- [8] M. Gebremicael and K. Tomsovic, "Initial Studies of Power Plant Construction under a Deregulated West African Power Pool using System Dynamics," *Proceedings of the 6th International Conference on Power Systems Operation and Planning*, Praia, Cape Verde, May 2005.
- [9] J. W. Forrester, *Industrial Dynamics*, Pegasus Communications, Waltham, MA.
- [10] A. Ford, *Modeling the Environment*, Island Press.
- [11] J. Sterman, *Business Dynamics*, Irwin McGraw-Hill, 2000.
- [12] G. Coyle, *Management System Dynamics*, John Wiley, 1977.
- [13] A. Dimitrovski, M. Gebremicael, K. Tomsovic, A. Ford and K. Vogstad, "Comprehensive Long Term Modeling of the Dynamics of Investment and Growth in Electric Power Systems," *2004 EPNES Workshop*, Mayaguez, Puerto Rico, July 13-14 2004.
- [14] A. Bose, K. Casavant, A. Dimitrovski, A. Ford, K. Tomsovic and L. Lutzenhiser, "Modeling the Interaction Between the Technical, Social, Economic and Environmental Components of Large Scale Electric Power Systems," *2003 EPNES Workshop*, Orlando, FL, Oct. 23-24 2003.
- [15] F. C. Schweppe, M. C. Caramanis, R. D. Tabors, and R. E. Bohn, *Spot Pricing of Electricity*. Boston, MA: Kluwer, 1988
- [16] T. Orfanogianni, G. Gross, "A General Formulation for LMP Evaluation," *IEEE Transactions on Power Systems*, Vol. 22, No. 3, August 2007, pp. 1163-1173
- [17] M. Rivier and J. I. Perez-Arriaga, "Computation and Decomposition of Spot Prices for Transmission Pricing," *Proc. 11th PSC Conf.*, 1993.

VI. BIOGRAPHIES

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Kevin Tomsovic (F'07) received the B.S. degree in electrical engineering from Michigan Technological University, Houghton, in 1982 and the M.S. and Ph.D. degrees in electrical engineering from the University of Washington, Seattle, in 1984 and 1987, respectively. Currently, he is Head and CTI Professor of the Department of Electrical Engineering and Computer Science at University of Tennessee, Knoxville. Visiting University positions have included Boston University, Boston, MA; National Cheng Kung University, Tainan, Taiwan, R.O.C.; National Sun Yat-Sen University, Kaohsiung, Taiwan, R.O.C.; and the Royal Institute of Technology, Stockholm, Sweden. He was on the faculty of Washington State University from 1992-2008. He held the Advanced Technology for Electrical Energy Chair at Kumamoto University, Kumamoto, Japan, from 1999 to 2000 and was an NSF program director in the ECS division from 2004 to 2006.